

30.09.2024

For a beam with flat top intensity profile of width d_{beam} and a wire of (effective) width $d_{\text{wire}} \leq d_{\text{beam}}$ at the last lens prior to focusing, I obtain the following focus intensity profile in the direction x perpendicular to the wire; I is a normalization:

$$I(x) = I \frac{f(\mu)}{\sqrt{2\pi}} \left[\text{sinc}\left(\frac{x}{w_x} f(\mu)\right) - \mu \text{sinc}\left(\mu \frac{x}{w_x} f(\mu)\right) \right]^2,$$

with $\mu = \frac{d_{\text{wire}}}{d_{\text{beam}}}$ and $f(\mu) = \sqrt{6(1 - e^{-1}) \frac{1 - \mu}{1 - \mu^3}}$; w_x is the $1/e^2$ waist of the central focus peak.

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The waist is related to the focal length F , beam diameter D at the aperture/lense and its wavelength λ as

$$w_x = \frac{F}{D} \frac{f(\mu)}{\pi} \lambda.$$

10.10.2024

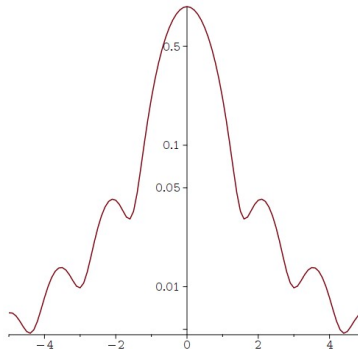
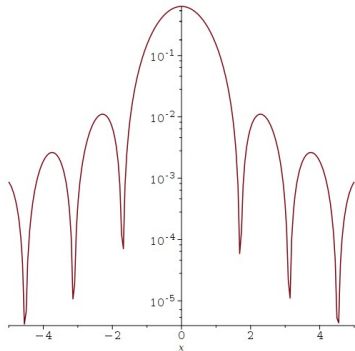
For a circularly symmetric beam with flat top intensity profile of diameter d_{beam} and **no hole** at the last lens prior to focusing, I obtain the following focus intensity profile as function of the radial direction $r = \sqrt{x^2 + y^2}$; I is a normalization:

$$I(x, y) = 2I \left[\frac{J_1 \left(\sqrt{2(1 - e^{-1})} \frac{2r}{w_0} \right)}{\frac{2r}{w_0}} \right]^2,$$

where w_0 is the $1/e^2$ waist of the central focus peak; $J_1(\cdot)$ is the Bessel function of the first kind.

$I(x, 0)$ for $I = w_0 = 1$: $\int_{-\infty}^{\infty} dy I(x, y)$ for $I = w_0 = 1$:

11.10.2024



[241011_FocusProfilePlot_->Pooyan.mw]